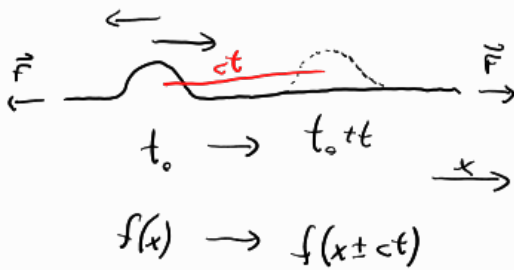


- prenos motnje po prožni snovi

potovanje motnje:

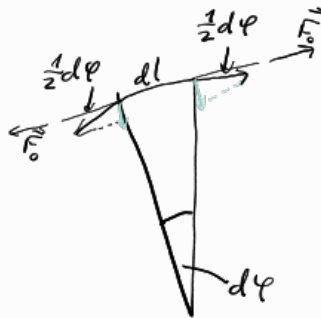
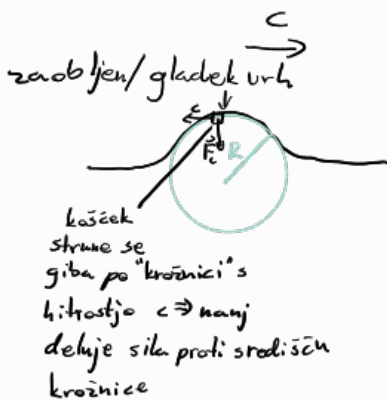


odmiki motenj se seštevajo

⇒ črta je linearna

- vztrajnost (ρ) → Newton

- prožnost (ϵ) → Hooke



$$dF_y = 2F_0 \frac{d\phi}{2}$$

$$dF_y = dF_c = dm a_c = dm \frac{c^2}{R}$$

$$dm = \frac{m}{l} \cdot dl = \frac{m}{l} \cdot R d\phi$$

$$dF_y = \frac{mc^2}{l} d\phi$$

motnja = odmik iz ravnovesne lege

1D $s(x, t)$

$s \leftarrow$ ni smiselno uporabiti x/y , ker obstajata prežno (transverzalno) in vzdolžno (longitudinalno) valovanje

$\vec{s} \perp \vec{z}$

$\vec{s} \parallel \vec{z}$

$$F_0 d\phi = \frac{mc^2}{l} d\phi$$

$$c^2 = \frac{F_0 \leftarrow \text{prožnost}}{m/l \leftarrow \text{vztrajnost}}$$

$$\frac{m}{l} = \frac{m \sqrt{S/l}}{l \cdot v} = \rho S$$

↑ dolžinska gostota

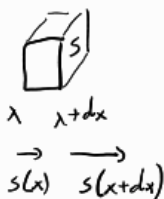
1D | strune

2D | vodna gladina, opne

3D | zvok

amplituda se zmanjša, ker je površina vedno večja (energija se ohranja) / volumen

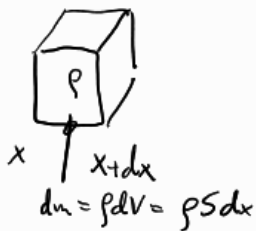
Hooke:



$$\frac{F}{S} = \epsilon \frac{s(x + dx) - s(x)}{dx} = \epsilon \frac{\partial s}{\partial x}$$

$\left(\frac{\partial s}{\partial x}\right)_{t=\text{konst.}}$ relativna deformacija

Newton:



$$dm \frac{\partial^2 s}{\partial t^2} = S \left(\frac{F}{S_{x+dx}} - \frac{F}{S_x} \right) = SE \left(\frac{\partial s}{\partial x}(x+dx) - \frac{\partial s}{\partial x}(x) \right)$$

$$\frac{\partial^2 s}{\partial t^2} \rho S = SE \frac{\partial^2 s}{\partial x^2}$$

$$\boxed{\frac{\partial^2 s}{\partial t^2} = \frac{E}{\rho} \frac{\partial^2 s}{\partial x^2}} \quad \text{Valovna enačba} \\ \text{-linearna}$$

$$s(x, t) = f(x \mp ct)$$

$$\frac{\partial f}{\partial t} = f'(x-ct)(-c)$$

$$\frac{\partial^2 f}{\partial t^2} = f''(x-ct)c^2$$

$$c^2 f''(x-ct) = \frac{E}{\rho} f''(x-ct)$$

$$\frac{\partial f}{\partial x} = f'(x-ct)(1)$$

$$\frac{\partial^2 f}{\partial x^2} = f''(x-ct)$$

$$\boxed{c^2 = \frac{E}{\rho}}$$

za longitudinalno

$$\boxed{c^2 = \frac{G}{\rho}}$$

za transverzalno

v tekočinah:

samo longitudinalno

$$\boxed{c^2 = \frac{1}{\rho \chi}}$$

v plinih:

$$\boxed{c^2 = \frac{\gamma RT}{M}}$$

zvok

v zraku 350 m/s

v vodi 2 km/s

v jeklu $4,5 \text{ km/s}$

.10

sinusno pot. val.

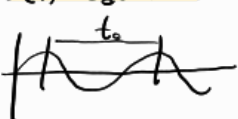
$t = \text{konst. (npr. fotografija)}$

$$s(x) = s_0 \sin kx$$



$x = \text{konst. (npr. boja na morju)}$

$$s(t) = s_0 \sin \omega t$$



$$s(x, t) = A \sin(k(x \mp ct)) \quad \omega t = kct$$

$$k = \frac{\omega}{c}$$

$$2\pi \frac{t}{T} = \omega t$$

k ... valovni vektor

$$\lambda = 2\pi$$

$$\rightarrow k_x = 2\pi \frac{x}{\lambda}$$

$$s(x, t) = A \sin(kx \mp \omega t + \phi) = A \sin\left(2\pi\left(\frac{x}{\lambda} - \frac{t}{T}\right)\right)$$

$$c = \frac{\omega}{k} = \frac{2\pi \nu \lambda}{2\pi} = \lambda \nu$$

Energija valovanja

$$W \rightarrow w = \frac{dW}{dV} \text{ energijska gostota}$$

gibanje snovi $v = \left(\frac{ds}{dt}\right)_{x=\text{konst.}} \rightarrow w_k = \frac{dW_k}{dV} = \frac{1}{2} \frac{dm}{dV} v^2 = \frac{1}{2} \rho v^2 = \frac{1}{2} \rho \left(\frac{ds}{dt}\right)^2$

$$W_{pr} = \frac{1}{2} kx^2$$

$$F = -kx \quad F = -\Delta \epsilon$$

$$W_{pr} = \frac{1}{2} \frac{\Delta \epsilon}{l} x^2$$

$$w_{pr} = \frac{W_{pr}}{V} = \frac{W}{Sl} = \frac{1}{2} \epsilon \left(\frac{x}{l}\right)^2 = \frac{1}{2} \epsilon \left(\frac{\partial s}{\partial x}\right)^2$$

$$\frac{\partial s}{\partial t} = (-c) f'$$

$$\frac{\partial s}{\partial x} = f'$$

$$w_k = \frac{1}{2} \rho c^2 (f')^2$$

$$w_{pr} = \frac{1}{2} \epsilon (f')^2$$

potujoče sinusno valovanje:

$$s(x, t) = s_0 \sin(\omega t - kx + \delta)$$

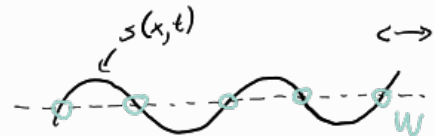
$$w_k = \frac{1}{2} \rho \omega^2 s_0^2 \cos^2(\omega t - kx + \delta)$$

$$w_{pr} = \frac{1}{2} \epsilon k^2 s_0^2 \cos^2(\omega t - kx + \delta)$$

$$\rho c^2 \frac{\omega^2}{c^2}$$

$$\delta \rho = \frac{1}{x} \frac{\partial s}{\partial x} = \rho c^2 \frac{\partial s}{\partial x}$$

$$\Rightarrow w_{pr} = \frac{(\delta \rho)^2}{2 \rho c^2}$$

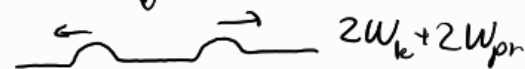
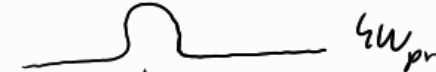
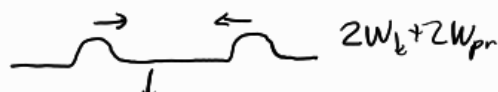
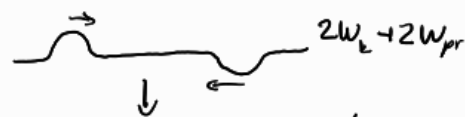


$\Rightarrow$ energija je v ravnovesni legi,
v ekstremih pa ne

Načelo superpozicije

če s_1, s_2 v istem prostoru in času

$$s = s_1 + s_2$$



$$s_1 = s_0 \sin(kx - \omega t)$$

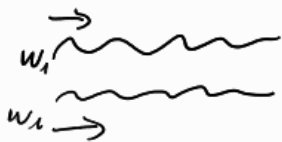
$$s_2 = s_0 \sin(kx - \omega t + \delta)$$

$$s = s_1 + s_2 = 2s_0 \sin(kx - \omega t + \frac{\delta}{2}) \cdot \cos \frac{\delta}{2}$$

$$A = 2s_0 \cos \frac{\delta}{2}$$

$$\max: \delta = 0, 2\pi, 4\pi, \dots$$

INTERFERENCA



$\rightarrow 2s_0$

$\downarrow$
 $4w_1 \times$ Energijski zakon ne velja

Ker ne moremo
hkrati ustrinjati
valovanja na istem
mestu z več razl. izviri

$$s_1 = s_0 \sin(kx - \omega t)$$

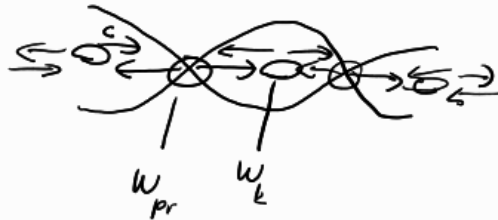
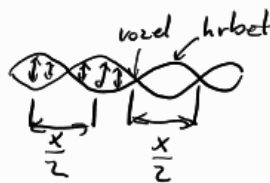
$$s_2 = s_0 \sin(kx + \omega t)$$

$$s = \underbrace{2s_0 \sin kx \cos \omega t}_{\text{stojече valovanje}}$$

$A(x)$

$$w_k = \frac{1}{2} \rho \left(\frac{\partial s}{\partial t} \right)^2 = \frac{1}{2} \rho A_0^2 \omega^2 \sin^2 kx \sin^2 \omega t$$

$$w_{pr} = \frac{1}{2} \rho c^2 \left(\frac{\partial s}{\partial x} \right)^2 = \frac{1}{2} \rho c^2 k^2 A_0^2 \cos^2 kx \cos^2 \omega t$$



w_{pr} in w_k se razporedi v obe smeri
(pol gre v levo, pol v desno) $\uparrow$ hitrost c

$$P = \frac{dW}{dt} = \frac{w dV}{dt} = \frac{w S c dt}{dt}$$

energijski tok $\boxed{j = \frac{P}{S} = w c} \left[\frac{W}{m^2} \right]$

gostota energijskega toka

$$\vec{j} = w \vec{c} = (w_k + w_{pr}) \vec{c} = \rho \omega^2 s_0^2 \cos^2(\omega t - kx) \cdot \vec{c}$$

dobimo "paketke" energije

$$\rightarrow \vec{j} = \rho \omega^2 s_0^2 \overline{\cos^2(\omega t - kx)} \cdot \vec{c} = \boxed{\frac{1}{2} \rho \omega^2 s_0^2 \cdot \vec{c}}$$

odvisno od gostote medija

$$\overline{\cos^2 x} = \overline{\sin^2 x}$$

$$2 \overline{\cos^2 x} = \overline{\sin^2 x} + \overline{\cos^2 x} = 1$$

$$\overline{\cos^2 x} = \frac{1}{2}$$

$$\left. \begin{aligned} \bar{w}_k^t &= \frac{1}{4} \rho A_0 \omega^2 \sin^2 kx \\ \bar{w}_{pr}^t &= \frac{1}{4} \rho A_0^2 \omega^2 \cos^2 kx \end{aligned} \right\} \bar{w}^t = \frac{1}{4} \rho A_0^2 \omega^2 \quad \text{ed kje pride } \frac{1}{2}?$$

$$\bar{w}_1 \propto \frac{1}{2} s_0^2 \quad \bar{w}_1$$

$$\bar{w} = 2\bar{w}_1 \propto 2 \cdot \frac{1}{2} s_0^2 = \left(\frac{A_0}{2}\right)^2 = \underline{\underline{\frac{1}{4} A_0^2}}$$

$$\delta_t = \frac{2l}{c} \quad s = \sum_{j=1}^N s_0 \sin(kx - \omega t + \delta_j)$$

$$\delta = \omega \delta_t = \omega \frac{2l}{c} = k \delta_x = k 2l$$

$$s = \sum s_0 \sin(kx - \omega t + (j-1)k \cdot 2l)$$

če so δ_j naključni $s=0$

če so δ_j celi mnogokratniki periode $\rightarrow s \neq 0$

$$\delta = k 2l = N \cdot 2\pi$$

lastno nihanje

$N=1$: osnovno last. nih. $\rightarrow v_0 = \frac{c}{\lambda} = \frac{c}{2l}$

$N=2$: višje harmonike

Piščal:

$N=1$:



odprta



polodprta



$$v_0 = \frac{c}{4l}$$

- ima samo lihe višje harmonike

$N=3$:

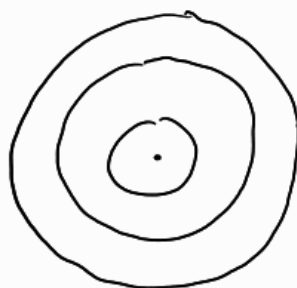


$$v_n = (2n+1)v_0$$

- imajo oster zvok (upr. orgle)

3D

$$s(\vec{r}, t)$$



vrh vala
valovna fronta

$$s = \text{konst.}$$

povezane točke z enako fazo

1D - točke
2D - krivulje
3D - ploskve

